

POC

Jira → GitLab Sync Service

Automated replication of Jira Stories, Bugs & Tasks to GitLab milestones
using a .NET 8 Windows Service — Proof of Concept

.NET 8

Jira REST API v3

GitLab REST API v4

Windows Service

Problem Statement

Three challenges driving the need for this sync service

01

Limited Jira Access

Entire team cannot see work items

Stories and Bugs are created in Jira, but only a few members have licences. Developers and QA who work in GitLab daily have zero visibility into sprint scope, ticket details, or planned work — creating a blind spot across the team.

02

Disconnected Tools

Code in GitLab, work tracking in Jira

All code repositories live in GitLab which the whole team can access, but sprint planning and stories remain siloed in Jira. Context-switching between two systems makes it hard to link commits to requirements or track sprint progress.

03

Manual Excel Tracking

Team managing tasks and status in spreadsheets

Without a shared system, members maintain tasks, assignments and work status in Excel files. This causes version conflicts, stale data, no sprint-wise tracking, and zero traceability between work items and the code being delivered.

Solution Benefits

What the team gains by syncing Jira to GitLab



Whole Team Visibility

Every member — dev, QA, designer — views sprint tickets in GitLab with no Jira licence needed.



Single Source of Truth

Replaces fragmented Excel sheets. Work status, assignees and story points always up to date.



Sprint-wise Tracking

Tickets map to GitLab milestones. Track velocity, completion rate and sprint health.



Code ↔ Work Traceability

Developers reference ticket IDs in commits — linking code changes to business requirements.



Zero Manual Effort

Runs automatically on schedule. No copy-pasting, no spreadsheet updates — fully hands-free.



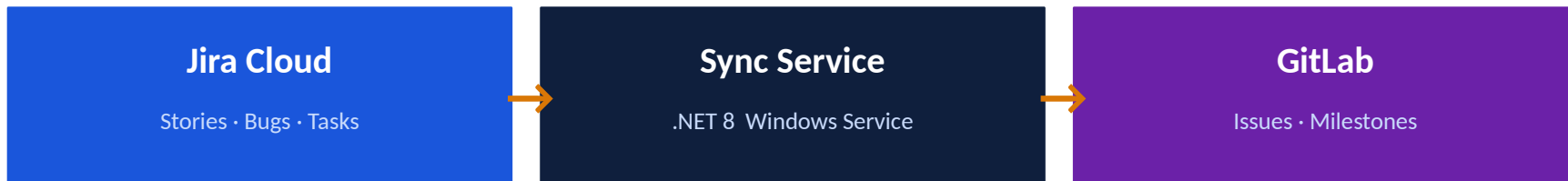
Better Retrospectives

GitLab burndown charts and milestone reports give accurate sprint data for stakeholder updates.

✓ No Jira licence needed ✓ Replaces Excel tracking ✓ Sprint progress in GitLab ✓ Fully automated

Solution Overview

A .NET 8 Windows Service that automatically replicates Jira tickets to GitLab on a configurable schedule



Scheduled Sync

Runs at configured times (e.g. 09:00 IST). Auto-retry on failure.



Smart Routing

Active milestone today → sprint. No milestone → Backlog.



Create & Update

New tickets created; existing issues updated and moved to correct milestone.



Field Mapping

ID, Description, Assignee, Sprint Dates, Story Points all synced.



Error Handling

Global exception handler with per-ticket isolation — one failure won't stop others.

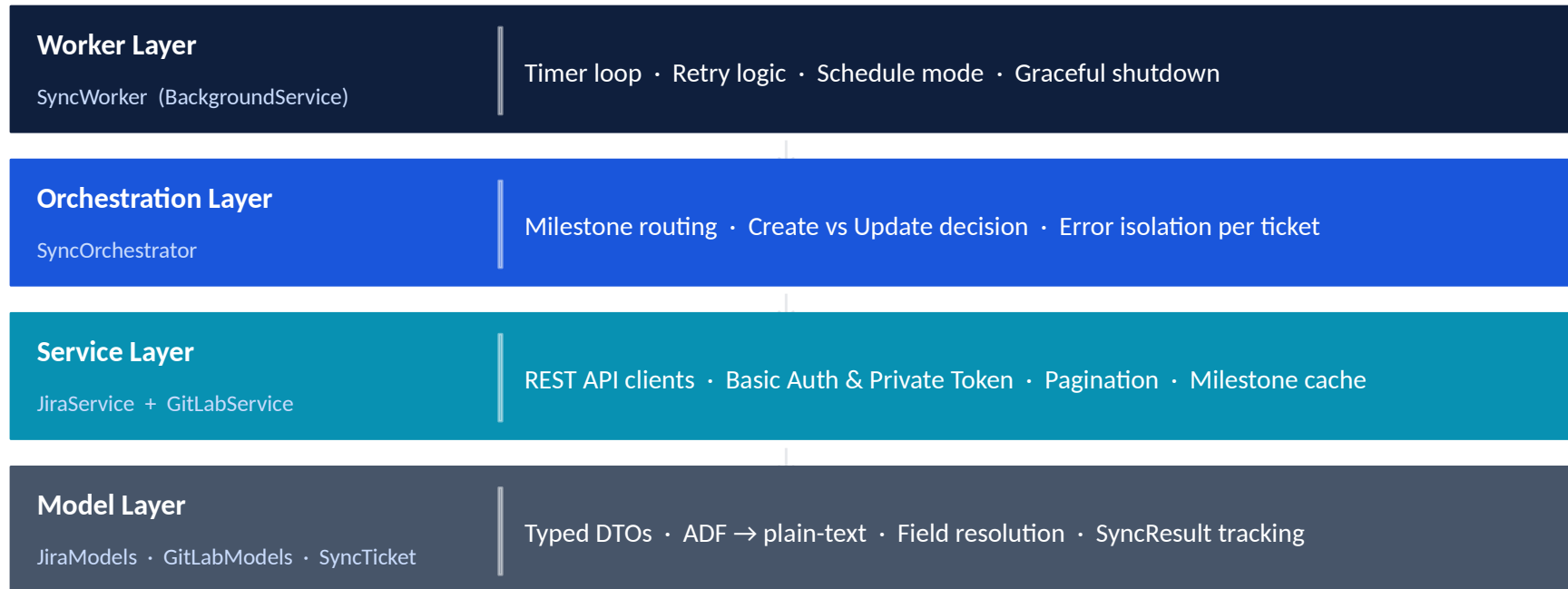


Config-Driven

All settings in appsettings.json — schedule, tokens, project key.

Architecture

Layered .NET 8 service with clean separation of concerns



Field Mapping

How Jira ticket data maps to GitLab issue fields

Jira Field	GitLab Field	Notes
Story / Bug / Task Key	Issue title prefix [PROJ-42]	Enables duplicate detection on re-sync
Summary	Issue title (after key prefix)	Full summary preserved
Description (ADF format)	Issue description (Markdown)	Atlassian ADF flattened to plain text
Assignee name / email	Assignee (matched by name/email)	Skipped on free-tier GitLab (403 on /users)
Sprint start → end date	Shown in description: From → To	Also drives milestone date matching
Story Points (customfield)	Issue weight field	Tries customfield_10016, 10028, story_points
Issue type (Story/Bug/Task)	GitLab label (story/bug/task)	Configurable via LabelMappings in config

Service Execution Log

Sync run showing 16 tickets created in GitLab Sprint-1 milestone

✓ Sync run complete — Created=16, Updated=0, Skipped=0, Failed=0

```
Received HTTP response headers after 505.3854ms - 200
Info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[101]
End processing HTTP request after 505.5256ms - 200
Info: JiraGitLabSync.Services.GitLabService[0]
Creating GitLab issue for SCRUM-5
Info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[100]
Start processing HTTP request POST https://gitlab.com/api/v4/projects/82304697/issues
Info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[100]
Sending HTTP request POST https://gitlab.com/api/v4/projects/82304697/issues
Info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[101]
Received HTTP response headers after 663.5037ms - 201
Info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[101]
End processing HTTP request after 663.5634ms - 201
Info: JiraGitLabSync.Services.SyncOrchestrator[0]
Created : [SCRUM-5] 'Screen is taking so much time to load and then timeout issue...' → milestone 'Sprint-1'
Info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[100]
Start processing HTTP request GET https://gitlab.com/api/v4/projects/82304697/issues?search=%5BSCRUM-4%5D&scope=all&per_page=5
Info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[100]
Sending HTTP request GET https://gitlab.com/api/v4/projects/82304697/issues?search=%5BSCRUM-4%5D&scope=all&per_page=5
Info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[101]
Received HTTP response headers after 408.6271ms - 200
Info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[101]
End processing HTTP request after 408.7168ms - 200
Info: JiraGitLabSync.Services.GitLabService[0]
Creating GitLab issue for SCRUM-4
Info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[100]
Start processing HTTP request POST https://gitlab.com/api/v4/projects/82304697/issues
Info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[100]
Sending HTTP request POST https://gitlab.com/api/v4/projects/82304697/issues
Info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[101]
Received HTTP response headers after 665.6504ms - 201
Info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[101]
End processing HTTP request after 665.7054ms - 201
Info: JiraGitLabSync.Services.SyncOrchestrator[0]
Created : [SCRUM-4] 'SQL Connection Timeout Error Occuring Occasionally. It happen...' → milestone 'Sprint-1'
Info: JiraGitLabSync.Services.SyncOrchestrator[0]
=== Sync run complete - Created=16, Updated=0, Skipped=0, Failed=0 ===
Info: JiraGitLabSync.Workers.SyncWorker[0]
Next sync in 15 minute(s) at ~14:06:54 UTC.
```

Service Execution Log

Subsequent run showing update sync — 1 created, 16 updated with no failures

✓ Sync run complete — Created=1, Updated=16, Skipped=0, Failed=0

```
End processing HTTP request after 585.8092ms - 200
info: JiraGitLabSync.Services.GitLabService[0]
      Updating GitLab issue !33 for SCRUM-4 → milestone_id=7433218
info: JiraGitLabSync.Services.GitLabService[0]
      PUT api/v4/projects/82304697/issues/33 body={"title":"[SCRUM-4] SQL Connection Timeout Error Occ
It happens every 2nd day in the evening 5PM IST time. ","description":"## Jira Reference: [SCRUM-4](
browse/SCRUM-4)\r\n\r\n**Type:** Story\r\n\r\n**Assignee:** Raghubir Singh\r\n\r\n**Sprint Period:** 2026-02-0
0\r\n\r\n**Priority:** Medium\r\n\r\n**Jira Status:** Done\r\n\r\n\r\n---\r\n\r\n\r\n### Description\r\n\r\n\r\nThis was a
JOB was running in the backend at 5:00 PM IST and occupying the DB resources. SQL Connection Timeout
this reason. As a solution, SQL Job is optimized and shifted and rescheduled to odd hours.\r\n\r\n","labo
y::medium","milestone_id":7433218,"assignee_ids":[],"weight":null}
info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[100]
      Start processing HTTP request PUT https://gitlab.com/api/v4/projects/82304697/issues/33
info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[100]
      Sending HTTP request PUT https://gitlab.com/api/v4/projects/82304697/issues/33
info: System.Net.Http.HttpClient.IGitLabService.ClientHandler[101]
      Received HTTP response headers after 1287.2385ms - 200
info: System.Net.Http.HttpClient.IGitLabService.LogicalHandler[101]
      End processing HTTP request after 1287.3595ms - 200
info: JiraGitLabSync.Services.GitLabService[0]
      PUT response body={"id":191949808,"iid":33,"project_id":82304697,"title":"[SCRUM-4] SQL Connect
curing Occasionally. It happens every 2nd day in the evening 5PM IST time.","description":"## Jira Re
https://gitlab.com/browse/SCRUM-4)\n\n**Type:** Story\n\n**Assignee:** Raghubir Singh\
info: JiraGitLabSync.Services.SyncOrchestrator[0]
      Updated : [SCRUM-4] !33 'SQL Connection Timeout Error Occuring Occasionally. It happen...' → miles
info: JiraGitLabSync.Services.SyncOrchestrator[0]
      === Sync run complete - Created=1, Updated=16, Skipped=0, Failed=0 ===
info: JiraGitLabSync.Workers.SyncWorker[0]
      Next sync in 15 minute(s) at ~16:42:20 UTC.
```


Source: Jira Board

4 To Do

3 In Progress

10 Done

The screenshot displays a Jira Board for the 'Enterprise-CRM' project. The board is organized into three columns: 'TO DO' (4 items), 'IN PROGRESS' (3 items), and 'DONE' (10 items). Each item is a card with a title, a date, and a status icon (e.g., a red 'X' for failed or a green checkmark for successful). The left sidebar shows the project's navigation menu, including 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', and 'Recent'. The top navigation bar includes a search bar and a '+ Create' button. The board's header shows the project name 'Enterprise-CRM' and a list of tabs: Summary, Timeline, Backlog, Board, Calendar, List, Forms, Goals, Development, Code, Archived work items, Docs, Shortcuts, and a plus sign for more options.

Spaces

Enterprise-CRM

Summary Timeline Backlog Board Calendar List Forms Goals Development Code Archived work items Docs Shortcuts

Search board Filter

TO DO 4

- 404 Page not found issue occurred while browsing URL
Apr 30, 2026
SCRUM-27
- 500 Internal Server Error - Failed to deserialize request payload.
May 27, 2026
SCRUM-32
- Authentication failed while logging into Customer Portal using valid credentials.
May 31, 2026
SCRUM-33
- Title: Add Product to Wishlist
Jun 9, 2026
SCRUM-36

IN PROGRESS 3

- Token-based authentication failing for integrated third-party application.
Jun 5, 2026
SCRUM-34
- User is not able to authenticate on login screen even providing the valid credentials?
May 28, 2026
SCRUM-35
- Duplicate ticket results displayed on UI for a single search due to incorrect API aggregation and database mapping
Apr 30, 2026
SCRUM-23

DONE 10

- SQL Connection Timeout Error Occuring Occasionally. It happens every 2nd day in the evening 5PM IST time.
Mar 6, 2026
SCRUM-4
- Screen is taking so much time to load and then timeout issue is occurring
Mar 5, 2026
SCRUM-5
- Validation on screen is not working properly on Chrome browser.
Mar 5, 2026
SCRUM-6
- Search results show incorrect resolution and missing source data on UI despite valid API response
Apr 30, 2026
SCRUM-22
- Email notification fails to send for ticket updates due to SMTP configuration issue
Apr 30, 2026

Result: Tickets Replicated to GitLab Sprint-1

17 Work Items

3 Labels

0 Failed

The screenshot displays the GitLab web interface for a project named 'Test-Project'. The browser address bar shows the URL: `gitlab.com/singh.raghubir2311-group/Test-Project/-/milestones/1#tab-issues`. The left sidebar contains a navigation menu with options like 'Test-Project', 'Learn GitLab', 'Pinned', 'Work items' (17), 'Merge requests' (0), 'Manage', 'Plan', 'Issue boards', 'Milestones' (selected), 'Wiki', 'AI', 'Code', 'Build', 'Secure', 'Deploy', 'Operate', 'Monitor', 'Analyze', 'Settings', 'Upgrade subscription', 'Help', and 'Collapse sidebar'. The main content area shows a milestone titled 'Sprint-1' for the period 'Feb 1, 2026–Jun 30, 2026'. Below the title, there are tabs for 'Work Items' (17), 'Merge requests' (0), 'Participants' (0), and 'Labels' (3). The 'Work Items' tab is active, showing a list of tickets under the 'Unstarted' (open and unassigned) category. The tickets include details such as title, ID, priority, and type. For example, ticket #34 is titled '[SCRUM-36] Title: Add Product to Wishlist' with a medium priority and story type. The right sidebar shows a progress bar at '0% complete' and various metrics: 'Start date Feb 1, 2026', 'Due date Jun 30, 2026 (23 days remaining)', 'Work items Open: 17 Closed: 0', 'Merge requests Open: 0 Closed: 0 Merged: 0', and 'Releases None'. A reference link 'singh.raghubir2311-...' is also visible at the bottom of the right sidebar.

Authentication Setup

How the service authenticates with Jira and GitLab

Jira — API Token (Basic Auth)

- 01 Go to id.atlassian.com → Security → API tokens
- 02 Click Create API token — give it a name
- 03 Copy the token (shown only once)
- 04 Set Username + ApiToken in appsettings.json
- 05 Auth header: Basic base64(email:token)

GitLab — Personal Access Token

- 01 Go to gitlab.com/-/user_settings/personal_access_tokens
- 02 Click Add new token — name it JiraGitLabSync
- 03 Select scope: api (full read/write access)
- 04 Copy the generated token (starts with glpat-)
- 05 Set PrivateToken in appsettings.json

POC Complete

Jira → GitLab Sync Service

16+

Tickets Synced

100%

Automated

3-Level

Milestone Fallback

0

Manual Steps